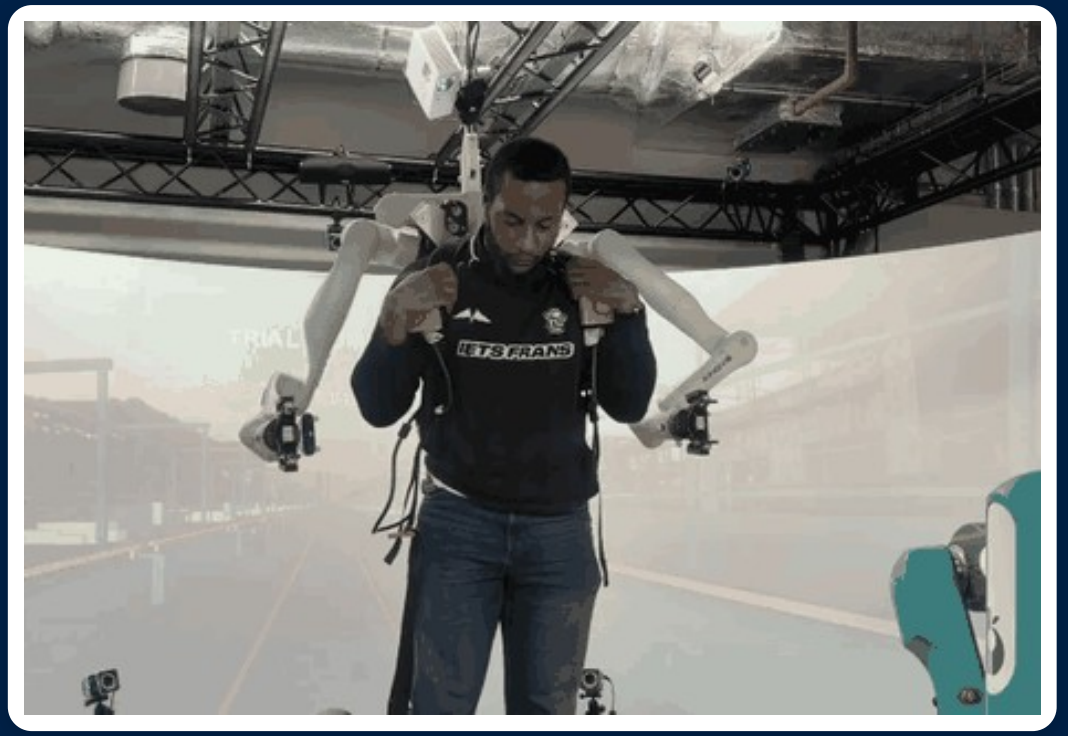
Imperial College London

World-Stable Supernumerary Effectors for Human Augmentation Under Locomotion-Scale Base Motion

Keeping a wearable robot's hands still while the wearer walks

Christian Akabueze
MSc Human and Biological Robotics
Imperial College London · MUVE Lab · 2026



Watch the hands, not the elbows.

Every step moves your extra hands



63–68 mm

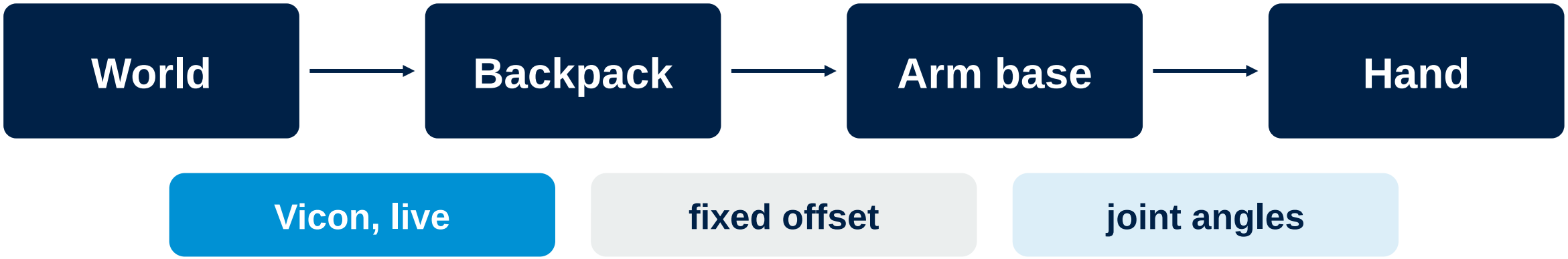
hand motion, arm locked (average)

~1.8 Hz

steps per second

Goal: hands fixed in the room, not the body

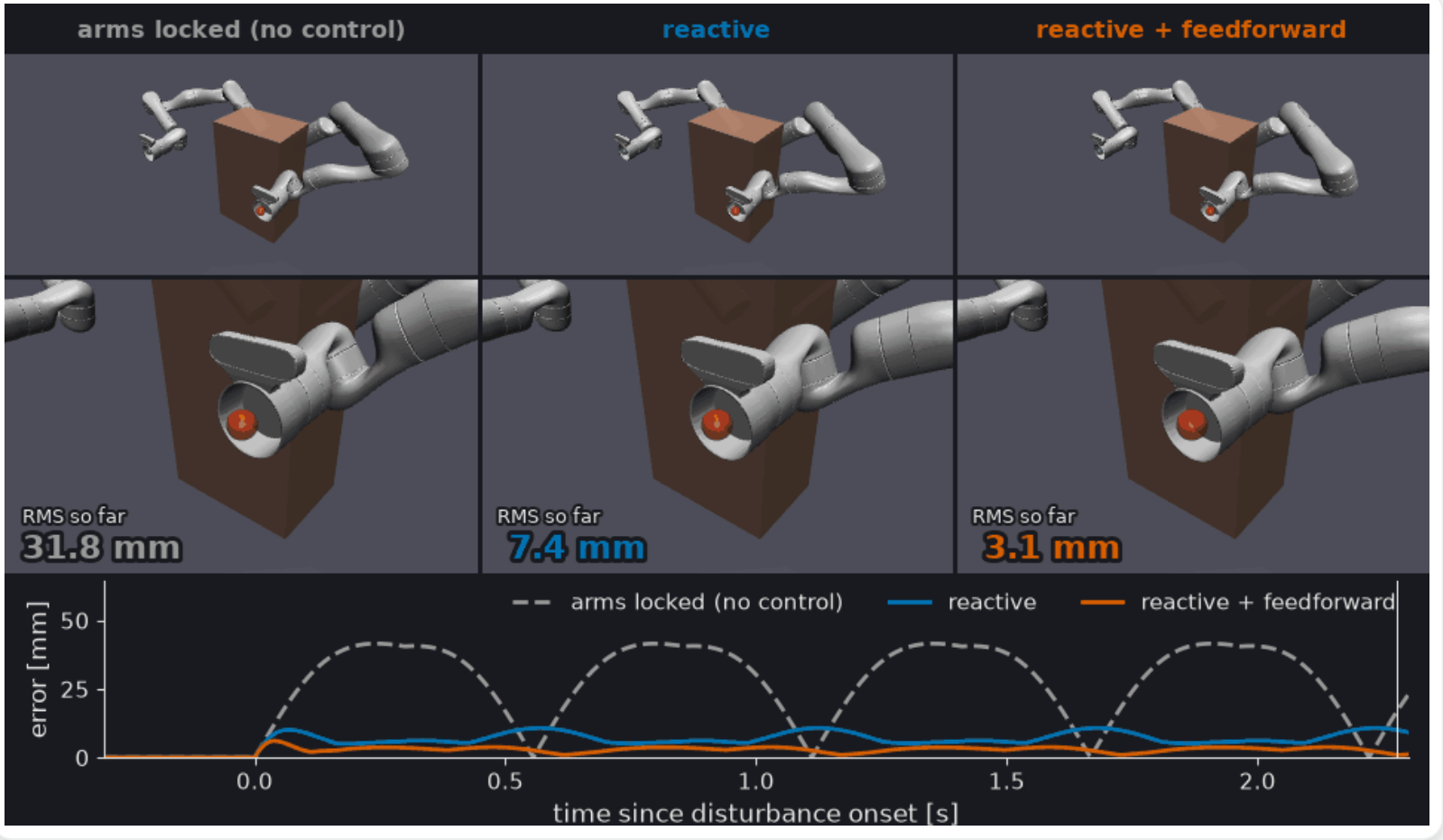
Measure the backpack, cancel it through the arm



$$\dot{q} = J^{+}_{\lambda}\,(K_p \cdot e + K_d \cdot \dot{e})$$

error in the room → joint speeds ▲ velocity term

Simulation: knowing the velocity is worth a lot



Hand error, 8 s average

31.0 mm

arms locked

6.7 mm

my controller

2.6 mm

+ speed fed forward

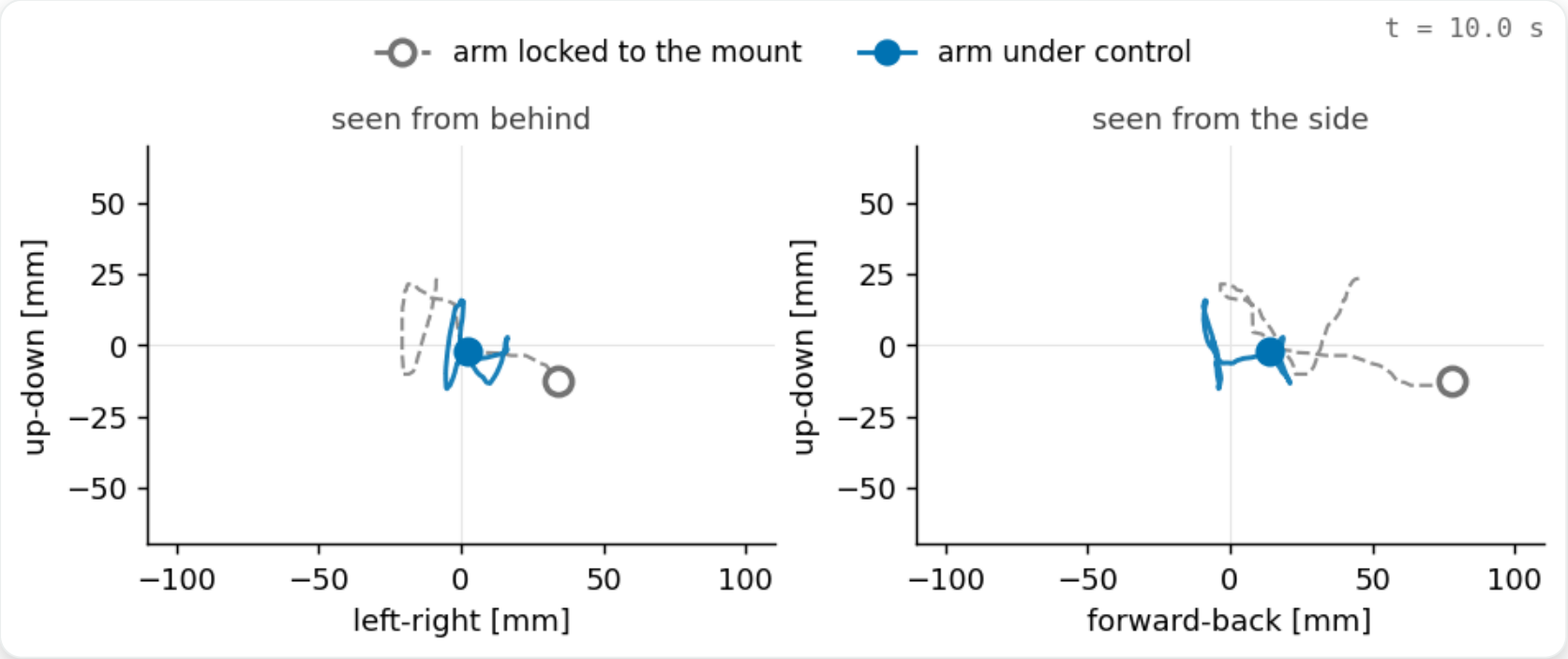
Making it usable: a filter that costs time



Smoothness costs time.

On real wearers: about 2/3 of walking motion removed

Controller: 400 Hz on the lab machine, no feedforward. Analysis: mine.



71%
0.5 m/s

63%
1.0 m/s

56%
1.5 m/s

Mean of 6 of 7 wearers · 128 trials

The obvious fix did almost nothing

One-wearer pilot: velocity gain 0.8 → 2.6. Hand motion near 1 Hz fell:

38%

predicted

9%

measured

Slow sway: 85% removed · Step bounce: only 29%

The velocity arrives 61–71 ms late



Velocity error: $2 \cdot \sin(\omega d/2)$

Late by 60 ms at 1.8 steps/s → **two-thirds** of the speed signal is wrong

Nothing fitted: explains the hardware better than a no-speed model, 6 of 6 wearers

What it achieved, and what comes next

~2/3

walking motion removed on hardware

Found the limit: a 61–71 ms delay

Next

1. Faster speed estimate
2. Then feedforward
   (sim: 6.7 → 2.6 mm)

Lesson: measure delays before tuning

Thank you. Questions?

Results, simulation, figures

github.com/fechachris4/msc_project

C++ real-time controller

github.com/fechachris4/HumanSL_MAIN

fecha412@gmail.com